Linduo Li

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EDUCATION

Institut Polytechnique de Paris (QS Ranking Worldwide: 41) — **Paris, France**
Master of Science, Data and Artificial Intelligence — **Expected 2027**

Northeastern University (US News Ranking in Engineering Programs: 34) — **Boston, MA, USA**
Master of Science, Data Analytics Engineering, GPA: 3.95/4.0 — **Dec. 2021**

Tongji University (QS Ranking in China: 8, QS Ranking Worldwide: 177) — **Shanghai, China**
Bachelor of Science, Geophysics, GPA: 85.6/100 — **Jun. 2018**

PUBLICATIONS

Journal Articles

- Liu, Z., Lin, G., Yan, Z., Li, L., Wu, X., Shi, J., ... & Wang, W. (2022). Predictive mutation signature of immunotherapy benefits in NSCLC based on machine learning algorithms. Frontiers in Immunology, 13, 989275. [link]
- Liang, H., Guo, Y., Chen, X., Ang, K. L., He, Y., Jiang, N., ... & Zhong, N. (2022). Artificial intelligence for stepwise diagnosis and monitoring of COVID-19. European radiology, 32(4), 2235-2245. [link]

Conference Proceedings

- Accessible AI for Archaeology: A User-Friendly Workflow for Predictive Site Detection.
  Computer Applications and Quantitative Methods in Archaeology (CAA), 2026. [oral presentation]
  Linduo Li, Yifan Wu, Zifeng Wang, Yongjie Zhuang.
- Fusing Text and Terrain: An LLM-Powered Pipeline for Preparing Archaeological Datasets from Literature and Remote Sensing Imagery. [link] [code]
  Computer Applications and Quantitative Methods in Archaeology - United Kingdom (CAA-UK), 2025. [oral presentation]
  Linduo Li, Yifan Wu, Zifeng Wang.

ACADEMIC EXPERIENCE

Information Retrieval for Tax Strategies (Advisor: Nils Holzenberger) — **Nov. 2025 – Mar. 2026**

- Constructed a legal knowledge base for tax authority retrieval by sampling and curating 3,055 documents from 24,626 U.S. tax authorities (IRC, Treasury Regulations, Revenue Rulings) using controlled noise injection and citation-aware deduplication
- Evaluated retrieval methods spanning sparse (BM25), dense (BGE-M3), hybrid (RRF), and query rewriting (decomposition) approaches, revealing persistent difficulty in identifying relevant authorities from goal-oriented queries
- Investigated reranking and relevance methods using instruction-aware neural rerankers and reasoning-based LLM judges, highlighting robustness–precision tradeoffs and limitations of relevance scoring in complex legal settings
- Designed and implemented corpus expansion pipelines that translate formal statutory language into practical tax-planning representations, adding structured fields such as trigger conditions, economic and legal effects to improve semantic match
- Conducted detailed retrieval failure analysis, identifying fundamental challenges such as the semantic–statutory vocabulary gap, multi-step reasoning invisibility, and cross-domain combination blindness
- Analyzed downstream LLM-based tax strategy generation under different authority conditions and performed perplexity-based evaluation, showing that imperfect retrieval can degrade generation quality relative to gold authorities

AI-Driven Dataset Pipeline for Archaeological Site Detection — **Oct. 2025 – Jan. 2026**

[pipeline code] [deployment code] [user interface] [write-up page]

- Developed an end-to-end archaeological site detection data preparation pipeline, transforming unstructured academic literature into machine-learning-ready remote sensing datasets for deep learning–based site discovery
- Designed a modular two-stage system combining LLM-based extraction of archaeological site coordinates from PDFs with automated generation of multi-channel satellite imagery datasets (Sentinel-2, SRTM-DEM) using Google Earth Engine
- Engineered scalable dataset generation workflows including positive sites, spatially informed negatives, land-cover negatives, and unlabeled samples, with geometric and radiometric augmentation to support robust model training
- Deployed an interactive web application integrating Google Gemini (multimodal reasoning), Google Earth Engine, and Google Maps, enabling researchers to upload papers, inspect extracted sites, visualize satellite context, and export structured datasets

Predictive Mutation Signature of Immunotherapy Benefits in NSCLC based on ML Algorithms — **Oct. 2021 – Mar. 2022**

- Assisted in designing the experimental workflow for a machine learning-based study on immunotherapy prediction tools, focusing on NSCLC patients using data from extensive cohorts
- Contributed to the development and application of XGBoost methods to identify key genomic features predictive of immunotherapy benefits from a dataset of 853 NSCLC patients

- Assisted in the internal and external validation of the ML-signature, demonstrating its enrichment in immune-related signaling pathways and its superiority in predictive value and stratification over tumor mutational burden (TMB)
- Participated in the analysis of ML-signature's predictive power in independent NSCLC patient cohorts treated with immune checkpoint inhibitors (ICI), confirming the signature's robustness and applicability

Artificial Intelligence for Stepwise Diagnosis and Monitoring of COVID-19 **Sep. 2020 – Apr. 2021**

- Contributed to the design of an experimental workflow for a novel artificial intelligence system utilizing deep learning and federated learning for COVID-19 diagnosis and patient monitoring
- Assisted in the development of deep learning methods for analyzing CT imaging, employing a stepwise diagnostic algorithm to accurately diagnose COVID-19 with or without clinical data
- Collaborated in the application of federated learning to refine independently built deep learning models, facilitating decentralized data processing while maintaining privacy and data protection
- Contributed to the successful integration of deep learning diagnostic models within a federated learning framework, demonstrating the model's adaptability and potential for global collaboration in healthcare

Research on Method of Machine Learning Applied to Surface Wave Suppression **Jan. 2018 – Jun. 2018**

- Developed advanced federated filtering algorithm with multiple features, innovatively generating machine learning labels for effective decoupling and noise reduction in seismic data
- Enhanced algorithm performance by implementing Support Vector Machine (SVM) model, achieving 3% improvement in classification accuracy while maintaining cost-effectiveness

PROFESSIONAL EXPERIENCE

BostonListener, LLM Developer, Boston MA USA **Jun. 2024 – Aug. 2025**

Uzei - AI-Powered Literature Review Platform (Python, JavaScript, HTML, CSS, LLMs, PostgreSQL)

- Architected and developed a literature review web application integrating 14+ AI models for automated academic research workflows, enabling researchers to process and analyze up to 120 papers per project with intelligent content extraction
- Designed AI-powered analysis pipeline using OpenRouter API and structured JSON schemas to automate metadata extraction, research question generation, and paper summarization, reducing manual literature review time from weeks to hours
- Built real-time streaming chat interface with comprehensive project context integration, allowing researchers to query their entire literature database through conversational AI that intelligently references analyzed papers, findings, and research gaps
- Developed Chrome browser extension with seamless one-click content extraction from academic publishers and PDFs, eliminating manual data entry processes and creating integrated workflow between web browsing and research documentation
- Implemented scalable multi-user collaboration platform with YAML-based project sharing, and automated BibTeX citation management, supporting concurrent research teams while maintaining optimal performance with optimized PostgreSQL [link]

Uchu - AI-Powered Parenting Guidance Platform (Python, JavaScript, HTML, CSS, LLMs, PostgreSQL)

- Architected and developed a comprehensive web-based AI consultation platform for parents, featuring real-time streaming responses, transparent AI reasoning visualization, and multi-step conversation flows
- Implemented a sophisticated state machine architecture with separated work and transition logic, enabling dynamic conversation pathways that adapt based on user inputs, search results, and contextual analysis
- Engineered a dual-window interface displaying both conversational content and AI thought processes simultaneously, enhancing user trust through unprecedented transparency in AI reasoning
- Developed internal and external knowledge search functionality integrating with Large Language Models to provide evidence-based parenting advice, allowing the system to reference both proprietary databases and current online resources
- Created a user profile management system that builds and maintains personalized consultation records, enabling the AI to tailor responses to specific family contexts and child development stages

AIYouCanSay, Founder & AI Engineer, Boston MA USA (Node.js, Python, LLMs, Azure) **Oct. 2023 – May 2024**

- Spearheaded the development of the forum platform AIYouCanSay, focusing exclusively on "question by human - answer by multiple Large Language Models (LLMs)," initiated a unique multi-bot response system
- Designed an innovative mode inviting external developers to contribute their own LLM APIs based on decentralization, fostering a collaborative and expansive bot ecosystem
- Developed a dynamic bot-selection feature, enabling users to choose or combine different specialized bots based on historical ratings for comprehensive and multi-perspective answers
- Integrated various LLM APIs, such as ChatGPT and Gemini, into a Discord-based test environment, demonstrating feasibility and effectiveness in real-time Q&A scenarios using Node.js and Python, deployed demo on Azure

Blend360, Data Scientist, Columbia MD USA **Aug. 2022 – Feb. 2023**

Retention Models for Top Dog Products Retail Client (PowerBI, GCP)

- Built and maintained data pipelines for processing 10-year customer subscription data on Google Cloud Platform, curated and optimized over 60 million dog product sales records from Sep. 2021 to Aug. 2022 for predictive analytics
- Collaborated with stakeholders on business requirements, built predictive models on customer churn using XGBoost, with a

classifier to identify potential churn customers in next 12 months and a regressor to generate monthly churn rate table

- Developed visualizations for customer churn predictions and inventory strategies using PowerBI, boosted profitability by $50,000 monthly via improving prediction accuracy by 2.48%, and reduced warehouse cost by around 5%
- Worked cross-functionally with client's software engineering team on model scalability and deployment, finished end-to-end documents from data collection and pipeline, to model building and finetuning, and to model deployment and maintenance

Attribution Models for Top Pet Products Retail Client (Snowflake, AWS)

- Conducted exploratory data analysis on over 150 million daily clickstream records, aggregated weekly user engagement metrics such as clickthrough and conversion rates at user action and merchandise category levels
- Coordinated with 21 key stakeholders across site merchandising, recommendation, and organic search teams to standardize metrics, leading to the creation of a new granularity table at widget and asset levels for attribution model development
- Utilized Plotly to visualize customer journeys, pinpointing key profitable user behaviors and creating the purchase funnel, developed linear model prototypes for ROI improvement analysis, laying groundwork for advanced attribution model building

Individual Projects

DualVectorFoil – AI Archaeology (GIS, Python, scikit-image) **May 2025 – Jun. 2025**

- Led development of interdisciplinary AI-powered archaeological detection pipeline using Python, GPT models, and remote sensing data, creating scalable methodology for identifying previously unknown archaeological sites in Amazon rainforest
- Built full-stack research platform with responsive web interface, enabling archaeologists and researchers worldwide to configure complex analysis parameters and execute multi-stage data processing workflows without technical expertise
- Engineered end-to-end data processing system handling deforestation patterns, satellite imagery, and elevation models, successfully identifying 5-10 validated archaeological candidates from 10,000+ initial sites through automated pattern recognition and AI interpretation
- Open-sourced scientifically rigorous framework in collaboration with archaeological and anthropological experts, providing comprehensive documentation and interactive visualizations that make advanced AI-driven heritage preservation tools accessible to global research community [write-up page]

Topic Modeling on 20 Newsgroups Dataset (Python, scikit-learn) **Apr. 2023 – Jun. 2023**

- Conducted document clustering and topic modeling on 20 Newsgroups Dataset, identifying latent semantic structures using a range of Natural Language Processing (NLP) algorithms
- Preprocessed text data through tokenization, lemmatization, and feature extraction using Term Frequency-Inverse Document Frequency for application in K-Means, Latent Dirichlet Allocation, and Non-negative Matrix Factorization models
- Trained and evaluated unsupervised learning models including K-Means Clustering, Latent Dirichlet Allocation, Non-negative Matrix Factorization, Top2Vec, and BERTopic, to discover latent topics and keywords for document classification
- Compared the performance of different topic modeling techniques, analyzing the effectiveness of clustering algorithms in capturing semantic groupings, and visualized the results of topic modeling for enhanced interpretation and further analysis

Boston Crime Analysis (Spark SQL) **Jan. 2022 – Mar. 2022**

- Conducted time series and spatial analysis using Spark SQL on a six-year crime dataset from Boston Police Department
- Compared time series trends of crime rates before and after COVID-19 pandemic, interpreted results with government policy, realized data-driven storytelling to inform public awareness
- Visualized variation in spatial distribution of different crime types from 2016 to 2021 using basemap package, revealing distinct crime patterns in various Boston regions, found that west and south Boston had higher crime rates than east and north Boston

Movie Recommendation Engine Development (Spark ML) **Aug. 2021 – Oct. 2021**

- Built data ETL pipeline to analyze movie rating dataset and conducted online analytical processing (OLAP) with Spark SQL
- Implemented Alternative Least Square (ALS) model to provide personalized movie recommendations
- Conducted model hyperparameters tuning with Spark ML cross-validation, and got the best model with root mean square error (RMSE) of predicted ratings smaller than 0.6 on over 10,000 rating entries

Skills

Programming Languages: Python, JavaScript, HTML, CSS, SQL
Data Science: Deep Learning, Machine Learning, Big Data, A/B Testing, ETL/ELT, Data Mining
Large Language Models: LLMs Finetuning, Prompt Engineering, Retrieval Augmented Generation
Cloud Platforms: Google Cloud Platform (GCP), Microsoft Azure, Amazon Web Services (AWS), Snowflake, Databricks

Volunteering Experience

Institute of Environment in Sustainable Development, Tongji University **Mar. 2015 – Jun. 2015**

- The International Student Conference on Environment and Sustainability (ISCES)
- Workshop on Planning for Greener Cities and South-South Cooperation in Africa